

- Explain the difference between codon optimization and codon harmonization
- Explain how tRNAs load the correct amino acids, corresponding to their anti-codon

Notes from the lecture

1. Introducing translation and the ribosome

Translation requires four key players: mRNA, tRNAs charged with amino acids, the ribosome (rRNA + proteins), and accessory factors for initiation, elongation, and termination.

Ribosomes consist of a small subunit that reads the mRNA and a large subunit that catalyzes peptide bond formation. They contain three functional binding sites: the A site (aminoacyl, entry for charged tRNA), the P site (peptidyl, holds growing chain), and the E site (exit, where uncharged tRNAs leave).

Ribosomal subunits are named after their sedimentation coefficients (Svedberg units): 30S + 50S = 70S in bacteria, and 40S + 60S = 80S in eukaryotes. rRNA is the catalytic core and evolves slowly, making it useful as an evolutionary marker.

tRNA serves as an adaptor, carrying specific amino acids and pairing codons with anticodons to ensure correct protein sequence.

1. Introducing translation and the ribosome (expanded and simplified)

Translation is the process by which the **genetic information stored in mRNA is converted into proteins**. Think of it as the cell's factory for making proteins. To do this, three key players must come together:

1. **mRNA** – the “blueprint” carrying codons (three-base words) that specify amino acids.
2. **tRNAs** – small adaptor molecules that “read” codons and bring the matching amino acid. Each has an **anticodon loop** that pairs with mRNA codons and an amino acid attached at the other end.
3. **Ribosome** – the molecular machine (made of rRNA + proteins) that coordinates the process.

The ribosome has **two subunits**:

- **Small subunit (30S in bacteria / 40S in eukaryotes)** → holds the mRNA and ensures correct codon–anticodon pairing.
- **Large subunit (50S / 60S)** → catalyzes the chemical reaction, forming peptide bonds between amino acids.

Inside the ribosome are **three functional sites** where tRNAs bind:

- **A site (Aminoacyl site):** Entry point for a new tRNA carrying the next amino acid.
- **P site (Peptidyl site):** Holds the growing polypeptide chain.

- **E site (Exit site):** Where the empty tRNA (no amino acid left) leaves the ribosome.

As the ribosome moves along the mRNA, it acts like a “conveyor belt”:

- A new tRNA enters the **A site**,
- the polypeptide chain is transferred from the **P site** to the new amino acid,
- the ribosome shifts forward, moving the tRNAs through the sites (P → E, A → P).

This cycle repeats until the protein is complete.

A clearer picture of what's going on

Think of the ribosome as a **3D printer for proteins**:

- The **mRNA is the instruction file** telling the printer which pieces to add.
 - The **tRNAs are the cartridges** that each carry the correct amino acid block.
 - The **ribosome** pulls the file (mRNA) through and clicks the amino acids together, one by one, until the full protein is printed.
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👉 In short: **Translation is the careful cooperation of mRNA, tRNA, and ribosomes. The ribosome “reads” mRNA and builds a protein by stringing amino acids together in the right order.**

2. Translation initiation

Initiation defines where translation begins, setting the reading frame. It differs in bacteria and eukaryotes but always requires initiator tRNAs, initiation factors, and ribosome assembly at the start codon.

2. Translation initiation (expanded and simplified)

Initiation is the first step of translation, and it's all about deciding **where to start reading the mRNA**. This is crucial because the ribosome reads mRNA in groups of three bases (codons). If it starts at the wrong place, the whole protein will be wrong (like misreading a sentence if you start at the wrong letter).

To get initiation right, the cell needs:

1. **Initiator tRNA** – a special tRNA that always brings the first amino acid. In bacteria, this is formyl-methionine (fMet); in eukaryotes, it's methionine (Met).

2. **Initiation factors (IFs or eIFs)** – proteins that help assemble everything in the right order and at the right spot.
 3. **Ribosomal subunits** – the small subunit first binds to the mRNA at the correct start site, then the large subunit joins to make a complete ribosome.
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A clearer picture of what's going on

Think of initiation as **finding the start of a recipe** in a cookbook:

- If you open the page and start in the middle, the dish won't make sense.
 - The **initiation factors are like bookmarks and helpers** that make sure you find the *real* start.
 - The **initiator tRNA is the first ingredient** — without it, you can't begin cooking.
 - Once the ribosome finds the start codon (AUG), it's like reading the first step of the recipe correctly. Then elongation (adding amino acids one by one) can begin smoothly.
-

👉 In short: **Initiation ensures the ribosome starts at the right codon, sets the correct reading frame, and places the first amino acid. Without proper initiation, the protein cannot be built correctly.**

2.1 Translation initiation in bacteria

The ribosome recognizes the **Shine–Dalgarno (SD) sequence** in the 5' UTR, a purine-rich sequence (AGGAGG) about 10 bases upstream of the start codon. It base-pairs with the 16S rRNA of the 30S subunit.

Start codons are usually **AUG**, but also **GUG** or **UUG**, and are decoded by a special initiator tRNA (**tRNA^{fMet}**). This methionine is formylated, which mimics a peptide bond and ensures initiation starts correctly. Elongator tRNAs cannot substitute for it.

Initiation factors **IF1, IF2, and IF3** coordinate binding of tRNA^{fMet} and assembly of the 50S subunit. GTP hydrolysis drives the process.

2.1 Translation initiation in bacteria (expanded and simplified)

In bacteria, the ribosome needs a signal to know **where to start** on the mRNA. That signal is the **Shine–Dalgarno sequence (SD sequence)**.

- The SD sequence is a short, purine-rich stretch of nucleotides (e.g., **AGGAGG**) in the 5' untranslated region (just before the coding region).

- It is located about 6–10 nucleotides upstream of the **start codon** (usually AUG, but sometimes GUG or UUG).
- The SD sequence pairs with a complementary region of the **16S rRNA** in the 30S ribosomal subunit. This base-pairing acts like Velcro, aligning the ribosome so the start codon is in the right place.

The first codon is decoded by a **special initiator tRNA (tRNA^{fMet})**. Unlike normal methionine tRNAs, this one carries a methionine that is chemically modified with a **formyl group (fMet)**. The formylation makes it look like the methionine is already part of a peptide bond. This “trick” ensures the ribosome only uses tRNA^{fMet} to start translation, and never in the middle of a protein.

Three **initiation factors** help the process:

- **IF1** helps tRNAs bind correctly.
- **IF2** delivers the initiator tRNA^{fMet} to the start codon using energy from GTP.
- **IF3** prevents the large 50S subunit from attaching too early.

Once the start codon is recognized and the initiator tRNA is in place, GTP is hydrolyzed (energy is spent), the 50S subunit joins, and the ribosome is ready to begin elongation.

A clearer picture of what's going on

Think of the Shine–Dalgarno sequence as the “**landing pad**” for the ribosome.

- The ribosome scans the mRNA until it finds the Shine–Dalgarno pad, where it can stick.
- This aligns the ribosome so the first codon (AUG) is perfectly in the window.
- Then, the special initiator tRNA^{fMet} walks in as the **first brick** of the protein wall.
- The initiation factors are like **construction workers** making sure everything lines up and the big machinery (50S subunit) doesn't move in too early.

 In short: **The Shine–Dalgarno sequence ensures the ribosome sits in the right spot, the initiator tRNA^{fMet} guarantees correct starting amino acid, and initiation factors coordinate the process until elongation can begin.**

2.2 Translation initiation in eukaryotes

Eukaryotic ribosomes recognize the **5' cap** with help from the **eIF4F complex** (eIF4E binds cap, eIF4G scaffolds, eIF4A unwinds mRNA). The 40S subunit, loaded with initiator Met-tRNAⁱ and eIF2–GTP, forms the **43S pre-initiation complex**. This scans the 5' UTR until it finds an AUG codon in a **Kozak sequence** (e.g., GCCRCCAUGG).

When AUG is recognized, eIF2 hydrolyzes GTP, factors are released, and the 60S subunit joins to form an 80S ribosome.

Eukaryotes can also initiate via **IRES (Internal Ribosome Entry Sites)**, bypassing the 5' cap, which accounts for ~10% of mRNAs.

Simplified:

In eukaryotes, ribosomes don't use Shine–Dalgarno sequences. Instead, they rely on the **5' cap** of the mRNA as the signal to find the start.

1. The small ribosomal subunit (40S) assembles with the initiator tRNA (carrying methionine, not formylated) and several initiation factors → this is called the **43S pre-initiation complex**.
2. At the 5' cap, special proteins (**eIF4E, eIF4G, eIF4A**) help the ribosome attach.
3. The ribosome then **scans along the mRNA** until it finds an AUG codon that is surrounded by a good **Kozak sequence** (consensus: GCC(A/G)CCAUGG).
4. Once the AUG is recognized, initiation factors are released, the large 60S subunit joins, and a complete 80S ribosome is formed, ready for elongation.

Eukaryotes can also use **Internal Ribosome Entry Sites (IRES)**, which allow ribosomes to bind in the middle of mRNAs, often used by viruses or in stress conditions.



Picture idea: Imagine the ribosome as a train. In eukaryotes, the train always enters the track at the **5' cap station**, then rolls forward until it finds the **“AUG start sign”** with the right Kozak decorations. Only then does the journey (translation) begin.

3. Elongation

After initiation, elongation is largely conserved between prokaryotes and eukaryotes.

1. An aminoacyl-tRNA enters the **A site** with help from **EF-Tu–GTP** (or eEF1A in eukaryotes). Hydrolysis of GTP ensures accuracy. EF-Ts regenerates EF-Tu–GTP.
2. The ribosome's peptidyl transferase (rRNA of the 50S subunit) catalyzes peptide bond formation, transferring the growing chain from the P site to the A site.
3. Translocation, powered by **EF-G–GTP** (or eEF2), moves the ribosome three nucleotides forward: the empty tRNA exits via the E site, the peptidyl-tRNA moves into the P site, and the A site becomes free.

This cycle repeats, extending the polypeptide.

Simplified:

Once initiation is complete, elongation begins — the ribosome adds amino acids one by one. The process repeats like a cycle:

1. **A site entry:** A charged tRNA (carrying the correct amino acid) enters the A site, helped by elongation factor **EF-Tu-GTP** (in bacteria) or **eEF1A** (in eukaryotes). Only the right anticodon-codon match is accepted.
2. **Peptide bond formation:** The ribosome's rRNA catalyzes a peptide bond between the amino acid in the A site and the growing chain in the P site. The chain is now attached to the new tRNA.
3. **Translocation:** The ribosome shifts forward one codon, powered by **EF-G-GTP** (bacteria) or **eEF2** (eukaryotes). The old tRNA moves to the E site and exits, the chain-tRNA moves to the P site, and the A site is open again.

 **Analogy:** Think of it as a **conveyor belt**:

- A new block (amino acid) arrives at the A site,
- it clicks onto the chain in the P site,
- then the belt shifts, moving everything over, making room for the next block.

4. Termination

Translation ends when a **stop codon (UAA, UAG, UGA)** enters the A site.

- In **bacteria**, RF1 recognizes UAA and UAG, RF2 recognizes UAA and UGA. RF3 recycles the factors.
- In **eukaryotes**, a single release factor, **eRF1**, recognizes all stop codons, assisted by eRF3.
Release factors catalyze hydrolysis of the bond between the peptide and the tRNA, releasing the polypeptide, after which the ribosome dissociates.

Simplified:

Translation ends when the ribosome encounters a **stop codon (UAA, UAG, UGA)**.

- No tRNAs match stop codons. Instead, **release factors** bind.
- In bacteria: RF1 recognizes UAA and UAG, RF2 recognizes UAA and UGA, and RF3 helps recycle.
- In eukaryotes: a single factor, **eRF1**, recognizes all stop codons with help from **eRF3**.

- The release factor triggers hydrolysis, cutting the polypeptide free from the tRNA. The ribosome then dissociates into its subunits.

 **Analogy:** Imagine the ribosome as a factory machine. Stop codons are like the “**end of recipe**” flag. Instead of a new block, a “release tool” (release factor) comes in and cuts the finished protein loose.

5. The genetic code

The genetic code is **universal and degenerate**: 61 codons encode 20 amino acids, while 3 codons are stops. Synonymous codons usually differ in the third base, which makes the code robust against mutations. Chemically similar amino acids also often share similar codons.

Simplified:

The genetic code is the “dictionary” that tells the ribosome which amino acid each codon represents.

- It is **universal** (almost all life uses the same code).
- It is **degenerate**: 61 codons specify only 20 amino acids. Most amino acids are coded by several codons.
- It is **non-overlapping** and read in triplets.

This redundancy makes the code robust: if a mutation changes one base, the amino acid may stay the same or change to a chemically similar one.

5.1 The wobble effect

The third base of a codon pairs less strictly with the first base of the anticodon. This allows one tRNA to recognize multiple codons. For example, the base inosine (I) in tRNA can pair with U or A. The wobble effect reduces the number of tRNAs needed in the cell.

Simplified:

The cell doesn’t need 61 different tRNAs. Why? Because the **third position of the codon (the “wobble position”)** allows flexible base-pairing.

Example: one tRNA anticodon with inosine (I) can pair with multiple codons like UUU and UUC (both coding for phenylalanine).

 **Analogy:** Think of wobble like **universal plugs** — you don’t need a new cable for every socket, some adaptors can fit more than one.

5.2 Codon usage

Codon usage varies across species. Some codons are rare in certain organisms because the corresponding tRNAs are scarce, slowing translation. Rare codons can play roles in regulating translation and co-translational folding.

- **Codon optimization** replaces codons with those most frequently used in the host to increase protein yield.
- **Codon harmonization** matches the translation dynamics of the native organism by preserving rare codons where needed to allow proper folding.

Simplified:

Even though multiple codons can code for the same amino acid, not all codons are used equally in every organism. This is called **codon bias**.

- Codons that match abundant tRNAs are translated faster.
- Rare codons slow translation down, which can help proteins fold properly.

Two strategies in biotechnology:

- **Codon optimization:** Replace rare codons with common ones to increase protein production.
- **Codon harmonization:** Keep some rare codons to mimic natural translation speed and folding.



Analogy: Codon bias is like language dialects — different organisms “prefer” certain codons the way people prefer certain words, even though they mean the same thing.

5.3 tRNA charging

tRNAs are loaded with the correct amino acid by **aminoacyl-tRNA synthetases**, enzymes that recognize specific tRNAs via identity elements in their anticodon and acceptor stem. The process requires ATP and ensures the genetic code is translated faithfully.

Simplified:

tRNAs are matched with the right amino acids by enzymes called **aminoacyl-tRNA synthetases**.

- Each synthetase is specific for one amino acid and its matching tRNAs.
- The enzyme attaches the correct amino acid to the 3' end of the tRNA, using energy from ATP.
- This ensures that the codon–anticodon recognition leads to the correct amino acid being inserted in the protein.

 **Analogy:** Think of aminoacyl-tRNA synthetases as **quality-control workers** who load the right ingredient into each delivery truck (tRNA). If the wrong cargo is loaded, the whole recipe fails.

Molecular Biology Week 5: Regulation of gene expression in prokaryotes

Learning objectives

After today's lecture and exercises, a student who has met the objectives will be able to:

- Identify promoters, terminators, and operator sequences in a bacterial DNA sequence (consensus and degenerate sequences).
- Explain the molecular mechanisms of prokaryotic regulation of transcriptional initiation and leader termination (repression, activation, and attenuation), translation initiation, and protein modification.
 - Explain in your own words why genes need to be regulated
 - Identify how the K_m and the V_{max} of an enzyme can be modulated to respond to environmental stimuli
 - Identify archetypical DNA-binding domains and explain how they interact with DNA
 - Describe the difference between regulatory and structural gene
 - Explain and propose engineerability solutions to using alternative sigma factors as regulators at the level of transcription initiation
 - Explain the difference between repressor and activator in the context of prokaryotic gene expression regulation
 - Differentiate between cis and trans-regulatory elements
 - Describe in detail the lac operon and explain how it is used to induce recombinant protein expression
 - Design mutant experiments to identify the function of regulators

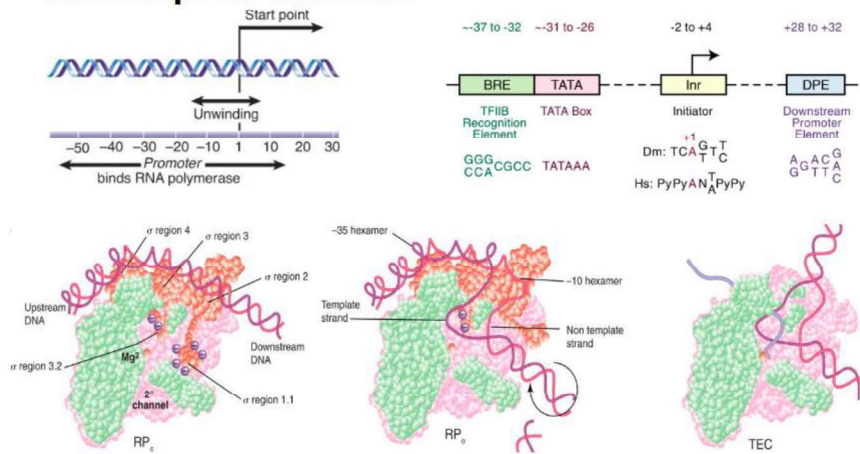
Notes from the lecture

1. Introduction

1.1 Why regulating gene expression?

Previous knowledge:

Previous knowledge: molecular details of transcription initiation



Order the elements according to the numbers in the image:

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Order the elements according to the numbers in the image

Correct order:

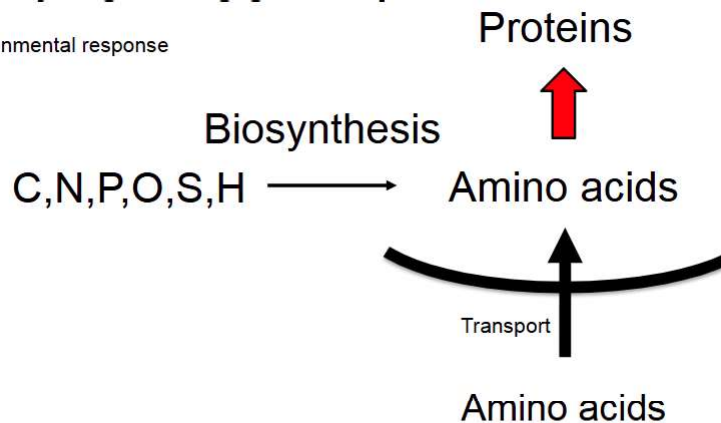
1. DNA
2. mRNA
3. Shine-dalgarno sequence
4. Start codon
5. -10 sequence
6. -35 sequence
7. Open reading frame (ORF)
8. Stop codon
9. Terminator
10. Leader

Correct responses: 59.02%

Why regulating gene expression?:

- “Economy” – a bacteria cannot do everything all the time because resources are limited
- “adaptation” – it has to do something in order to not die in a possible new environment
- “survival” the bacteria can regulate what kind of proteins that will be made (?) it makes room for “engineerability”

- Environmental response



Amino acids are very costly for bacteria. If it cannot “reach” the amino acids it cannot use it.

So what if the supply of amino acids is reduced? – amino acid biosynthesis must increase.

In this case in the central dogma some genes will be transcribed into mRNA and those mRNA’s will be translated into proteins. The transcription is the best way to make a change that is going to last.

- The Central Dogma



2. Regulation of enzyme activity

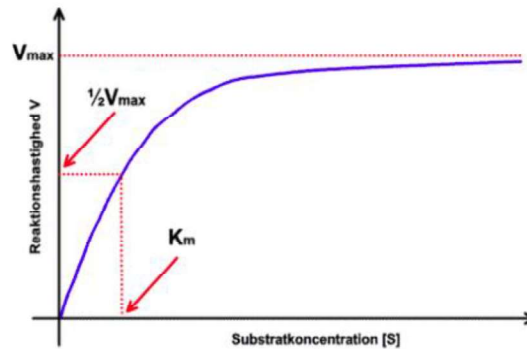
2. Regulation of enzyme activity

- Change K_M
 - Competitive inhibition (-)
 - Allosteric modulation (+/-)
- Change V_{max}
 - Allosteric modulation (+/-)
- Change the amount of enzyme (+/-)
 - **Regulate gene expression**

Fixing what the cell already have

Modifying how much you have

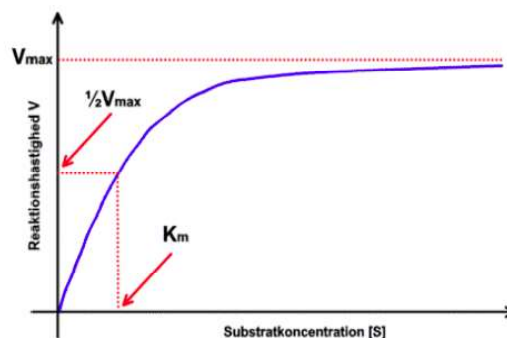
- Enzymes are protein catalyzers that transform substrates into products.
- The speed of the conversion depends on:
 - Substrate concentration
 - Enzyme concentration
 - pH, temperature
 - Other cofactors and modifications
- How can the cell change the flux through a pathway?
 - Increase the rate by which the enzymes convert substrates into products
 - Modify K_M and V_{max}



How can the cell change the flux through a pathway?: check in the enzyme concentration – but this is tricky. – else increase or decrease the rate by which the enzymes convert substrates into products.

K_M is the Michaelis-Menten constant – the higher the K_M the more substrate is needed to make the process run – a lower K_M is always gonna be more effective.

- How can the cell change the flux through a pathway?
 - Increase the rate by which the enzymes convert substrates into products
 - Modify K_M and V_{max}
 - Competitive inhibition
 - Allosteric control

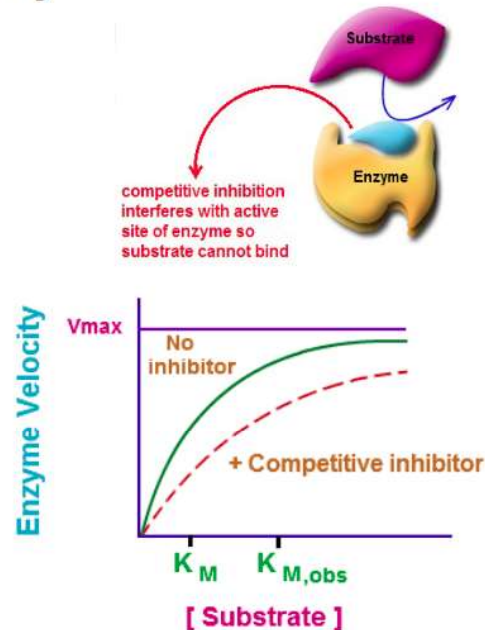


2.1 Competitive inhibition

2. Regulation of enzyme activity

2.1 Competitive inhibition

- Competitive inhibition occurs when an inhibitor molecule competes with the substrate for binding to the active site of the enzyme. Because both the substrate and the inhibitor are competing for the same site, the presence of a competitive inhibitor reduces the proportion of enzyme molecules bound to the substrate
- The abundance of available enzyme is reduced
- Therefore, the “apparent K_M ” is increased
- However, “ V_{max} ” remains unchanged, as enough substrate can outcompete the inhibitor



K_M is going to be increased when competitive inhibitions occurs – so more substrate is going to be needed for the process to occur.

The v_{max} is constant.

Both substrate and inhibitor want to bind to the active site

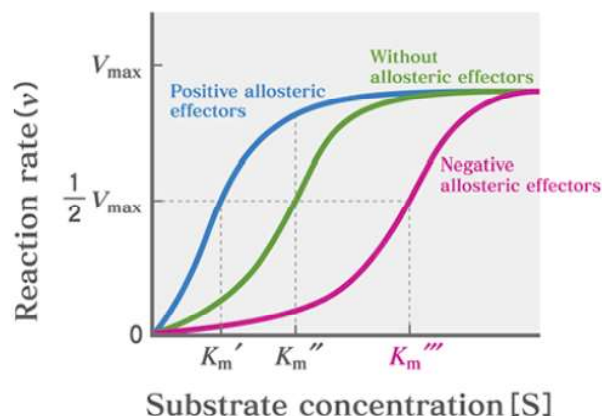
2.2 Allosteric control

Allosteric inhibition is something that binds to something on the enzyme that is not the active site but it affects that the enzyme is worse or better to bind the substrate – it can be either positively or negatively modulating.

2. Regulation of enzyme activity

2.2 Allosteric control

- Allosteric control refers to the regulation of an enzyme's activity by binding a molecule (an allosteric modulator or effector) to a site other than the enzyme's active site, called the allosteric site. This binding induces a conformational change in the enzyme, which can either increase (positive allosteric regulation) or decrease (negative allosteric regulation) the enzyme's activity.
- Allosteric control can modify both K_M and V_{max} positively or negatively

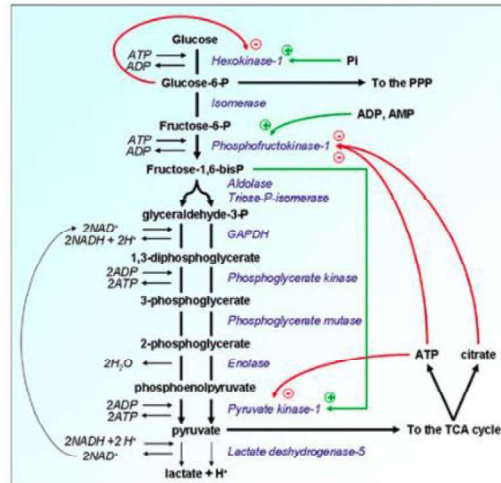


Example of allosteric control:

2. Regulation of enzyme activity

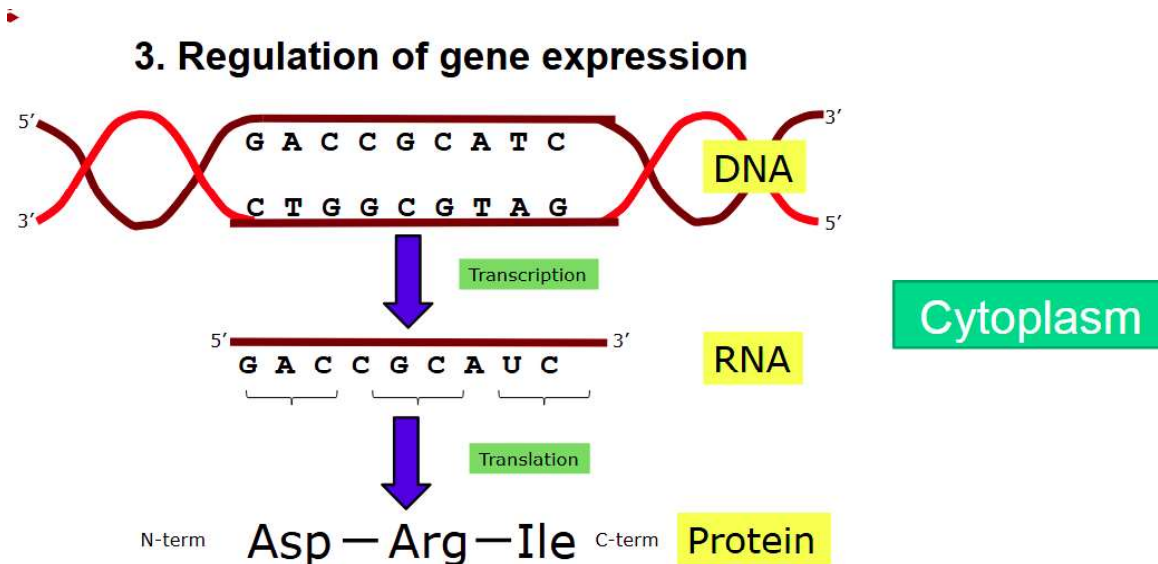
2.2 Allosteric control

- It is common to find metabolic routes that are allosterically regulated by the product.
- This mechanism avoids stops the production of a metabolite when such metabolite is produced in enough amount or the objective of the route is fulfilled (i.e., producing energy)



So to change the K_m we can change the allosteric control (modulation) or competitive inhibition – v_{max} can only be modified by allosteric modulation/control, for competitive inhibition v_{max} would remain the same.

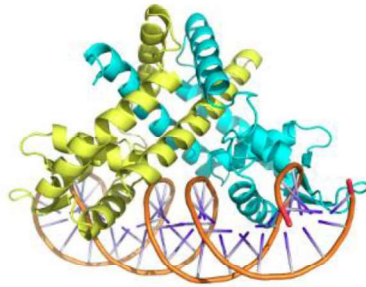
3. Regulation of gene expression



In this case everything happens in the cytoplasm (the change from transcription to the translation happens very fast – not like next week's stuff)

DNA binding proteins:

- To regulate the expression of a protein (enzyme) at the transcription level, there must be a connection between a regulatory agent (normally another protein) and the DNA.
- Proteins need a DNA Binding Domain (DBD) to interact with DNA
- A DNA binding domain (DBD) is a protein segment that allows it to bind to DNA. These domains often recognize and interact with specific sequences of DNA, enabling the protein to exert its regulatory effects.



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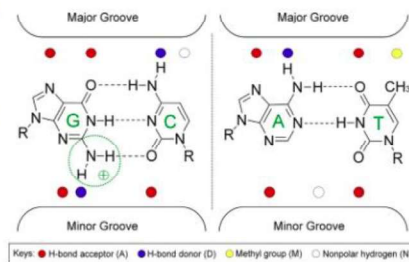
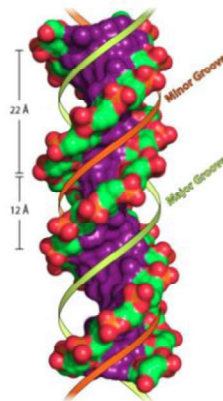
4. DNA-binding domains (DBDs)



4. DNA-binding domains

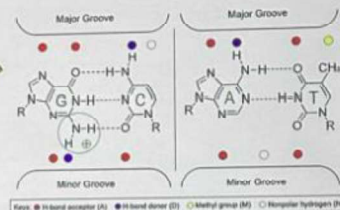
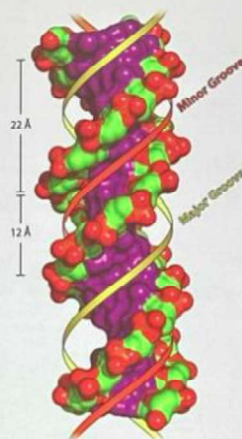
4.1 Major vs minor groove of B-DNA

- B-DNA is a defined helical structure defined by a major and a minor groove:
- **Major Groove Binding:** is wider and has more unique features that proteins can "read". Many DBDs interact with the major groove because it offers more distinct and accessible patterns for hydrogen bonding and other interactions.
- **Minor Groove Binding:** Some proteins and molecules can bind to the minor groove, but due to its narrower and less distinct nature, it's less common for sequence-specific DNA-protein interactions.



DNA binding domains

- DNA binding domains are parts of a protein individually folded that can interact with specific sequences of the DNA
- Activators and repressors always contain a DNA binding domain
- DNA binding domains usually interact with the major groove of DNA because the amount and diversity of chemical groups available is greater than in the minor groove.
- There are some known conserved DNA binding domains for example Leucine zippers, Zinc finger or Helix-turn-helix



Every single protein that wants to interact with the DNA needs a DBD DNA binding domain.

Most proteins will go through the major groove – because there is more space and more chemistry

4.1 Major vs Minor B-DNA groove

4.2 DBDs specificity

- The interactions made by a DNA binding protein to a specific site on the DNA is accomplished by specific contacts between approximately **five nucleotides** on the DNA and the recognition helix on the protein – IT IS TRUE but what's the problem?
- What is the probability for a specific 5 bp sequence in a genome?
- 0.09765% or about 1 in 1.024
- Human genome: 2.93 million times
- If 10: 1 in 286.110, If 15: 1 in 11.444

A longer sequence is going to have a less frequency – we need to make it even more specific – a single DNA binding protein will bind a 5 bp, but if it combines with another it needs to find 10 specific basepairs it gets way more specific and if that binds to yet another one it gets even more specific.

Increased specificity, palindromic sequences, stability and strength

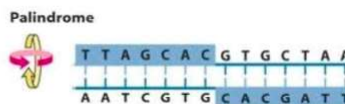


4. DNA-binding domains

4.2 DBDs specificity



- Many DNA-binding proteins function as dimers (pairs of protein molecules). There are several reasons for this:
- **Increased Specificity:** By working in pairs, proteins can recognize longer stretches of DNA, which enhances their binding specificity.
- **Palindromic Sequences:** When these dimers bind to DNA, they often recognize sequences that are palindromic, meaning they read the same forwards and backwards. This symmetry allows both protein subunits in the dimer to bind DNA in a similar manner.
- **Stability and Strength:** Two binding domains can offer a stronger, more stable interaction with DNA compared to a monomer.
- Eva, can I see bees in a cave?
- Was it Eliot's toilet I saw?
- Murder for a jar of red rum



So, they make the job way more efficient by forming dimers - Restriction enzymes are dimers.

(we remember the palindromic sequences from restriction enzymes)

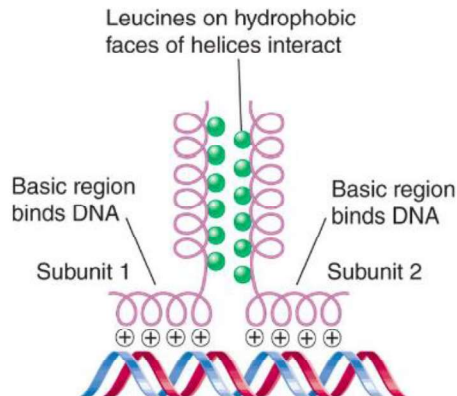
And they look more or less like the figure in the slide above.

4.3 Types of DBDs

4. DNA- binding domains

4.3 Types of DBDs

- Leucine Zippers
- The name comes from a periodic repetition of leucine residues at every seventh position in a segment of the protein
- These leucine residues are responsible for dimerization (joining two protein subunits). The "zipper" part of the protein forms a coiled-coil structure, and another region of the protein forms an alpha-helix that binds to the **major groove** of DNA.



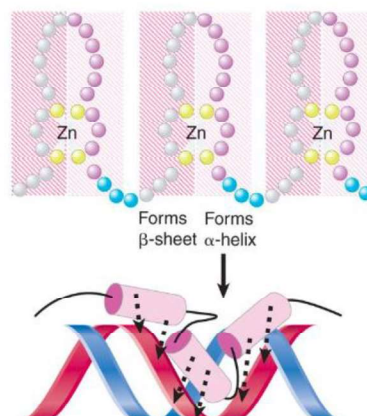
The name of the domains should be something we know and they usually interact with a domain of the DNA and we should know how and why. (The plusses are amino acids.)

- Leucine Zippers. (the major groove of DNA)
- Zinc Fingers (the major groove of DNA)
- Helix-turn-helic (both the major and the minor groove – super specific because of this)

4. DNA- binding domains

4.3 Types of DBDs

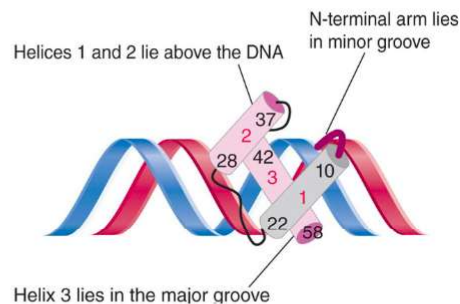
- Zinc Fingers
- This DBD bind single zinc ions
- These motifs commonly consist of a short beta-sheet and an alpha-helix, with the zinc ion stabilized by cysteine and/or histidine residues. The alpha-helix typically inserts into the **major groove** of DNA.



4. DNA- binding domains

4.3 Types of DBDs

- Helix-turn-helix (Helix-three)
- Three helices connected by loops
- The C-terminal helix (3) lies in the **major groove** of DNA and is the recognition helix, while the middle lies across the DNA (2) and the N-terminal domain lies in the **minor groove** (3)

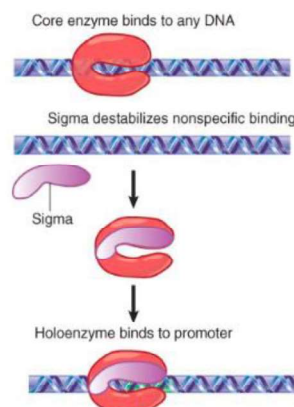


5. Regulation at the level of transcription Initiation

5. Regulation at the level of transcription initiation

5.1 Alternative Sigma factors

- Holoenzyme: core enzyme + Sigma factor
- RNA polymerase consists of multiple subunits. The core enzyme can bind to DNA and catalyze RNA synthesis, but it doesn't have specificity—it doesn't know where to start.
- The sigma factor (e.g., S70 in *E. coli* under normal growth conditions) is a separate protein that forms what is known as the RNA polymerase holoenzyme when associated with the core enzyme. This holoenzyme is capable of **specific promoter recognition** (-10 TATAAT- and -35 regions).



Alternative sigma factors^ - is going to recognize a new set of genes that is going to (?)

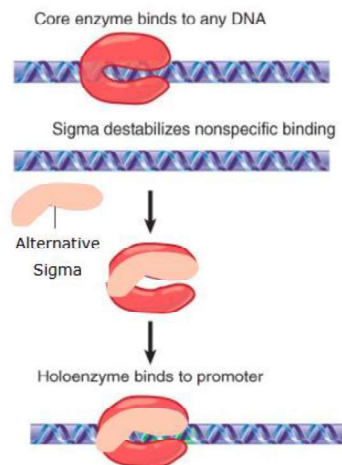
the bacteria has a way to get away from a heat shock (like fever) which is the alternative sigma factor.

5.1 Alternative sigma factors

5. Regulation at the level of transcription initiation

5.1 Alternative Sigma factors

- By changing the sigma factor, the preferences of the RNA polymerase is changed, enabling a total change in the global expression profile
- Sigma factors unlock the expression of sets of genes that respond to different stimuli.
- **Example:** Sigma 32 is associated with Heat Shock. Sigma 32 leads direct the RNA polymerase to heat shock promoters, leading to the expression of heat shock genes.



5. Regulation at the level of transcription initiation

5.1 Alternative Sigma factors

- *Streptomyces coelicolor* A3(2) contains 66 different sigma factors!!!
- By changing the sigma factor, the preferences of the RNA polymerase is changed, enabling a total change in the global expression profile.

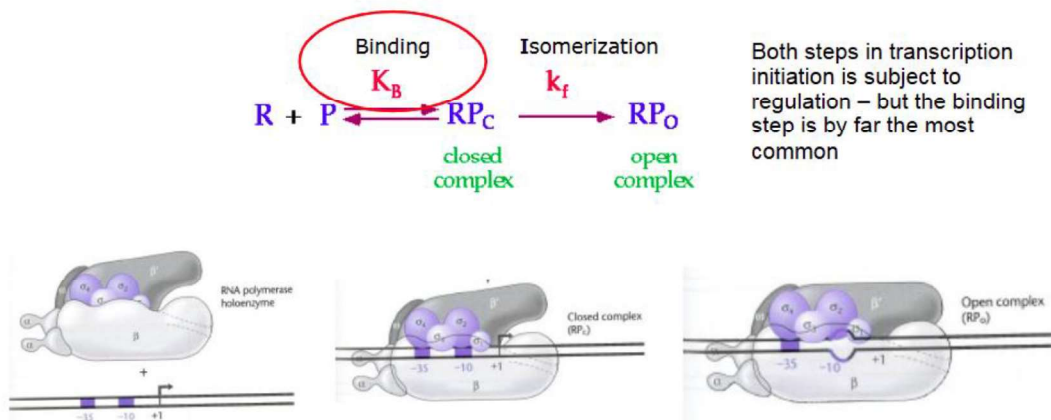
Gene	Factor	Use	-35 Sequence	Separation	-10 Sequence
<i>rpoD</i>	σ^{70}	general	TTGACA	16-18 bp	TATAAT
<i>rpoH</i>	σ^{32}	heat shock	CCCTTGAA	13-15 bp	CCCGATNT
<i>rpoE</i>	σ^E	heat shock	not known	not known	not known
<i>rpoN</i>	σ^{54}	nitrogen	CTGGNA	6 bp	TTGCA
<i>flaA</i>	σ^F	flagellar	CTAAA	15 bp	GCCGATAA



Regulation at the level of transcription initiation:

KB is going to be the limiting factor.

5. Regulation at the level of transcription initiation



We have to learn how to regulated K_B .

When an enzyme is regulated by competitive inhibition

The K_m and the V_{max} are reduced

0%

The K_m is reduced but the V_{max} remains unchanged

0%

The K_m remains unchanged and the V_{max} is reduced

0%

The K_m is increased but the V_{max} remains unchanged

0%

Answer: The K_m is increased but the V_{max} remains unchanged

Which of the following DNA binding domains interact with the major groove of B-DNA

Leucine zippers, zinc fingers, helix-turn-helix

0%

Isoleucine zippers, zinc fingers, helix-turn-helix

0%

Leucine zippers, beta-sheet bundle, helix-turn-helix

0%

Leucine zippers, cobalt fingers, helix-alpha-helix

0%

Answer: Leucine zippers, zinc fingers, helix-turn-helix

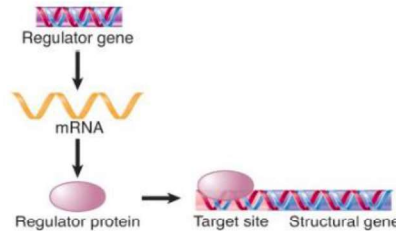
5.2 Structural vs regulatory gene

5. Regulation at the level of transcription initiation

5.3 Structural vs regulatory gene



- A **structural gene** – A gene that encodes any RNA or protein product other than a regulator.
- A **regulator gene** – A gene that encodes a product (typically protein) that controls the expression of other genes (usually at the level of transcription initiation).



A regulator genes role is to control the expression of other genes. (usually at Kb – the level of transcription initiation).

5.4 Repressors

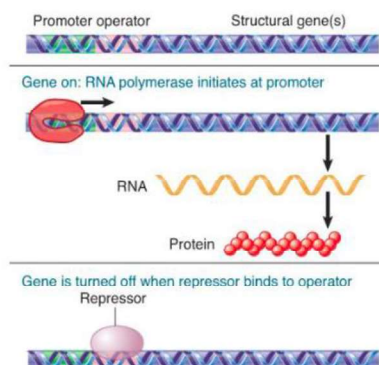
Negative regulation, repressor protein, operator:

5. Regulation at the level of transcription initiation

5.3 Repressor



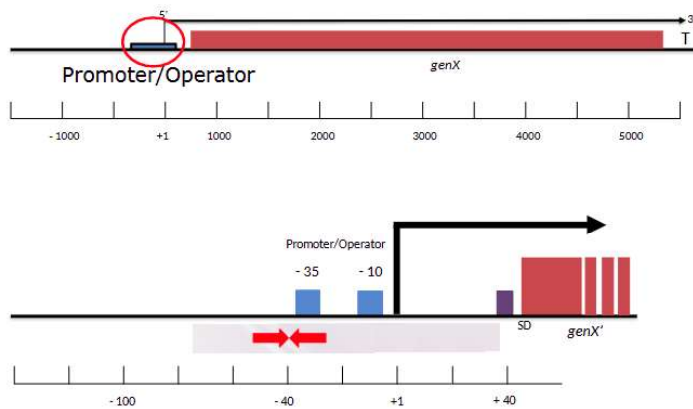
- When a gene is **ON by default**, and it has to stop, it is subjected to **negative regulation** by a **repressor protein** bound to the **operator**.
- **The operator overlaps with the RNA polymerase binding site**, avoiding RNA polymerase binding and preventing the gene from being expressed.



Protein that is ON by default and an operator that overlaps with the RNA polymerase binding site.

5. Regulation at the level of transcription initiation

5.3 Repressor



The promoter/operator will ALWAYS overlap with (whatever the red circle circles)

If the repressor binds there will be no transcription and therefore no translation.

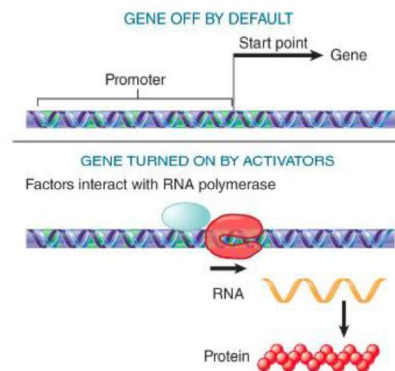
5.3 Activators

5. Regulation at the level of transcription initiation

5.3 Activator



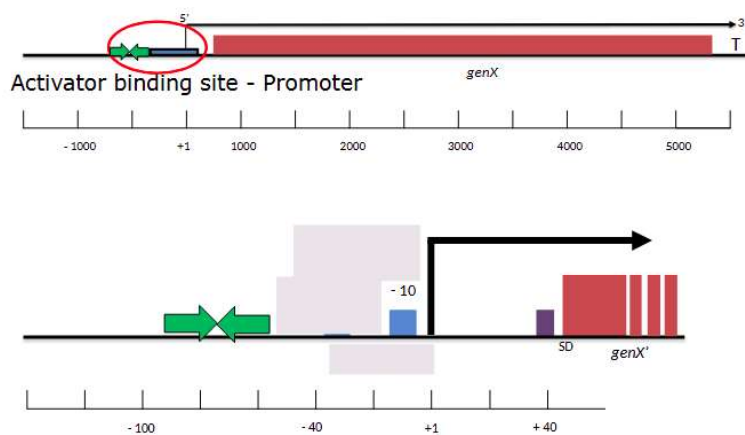
- When a gene is **OFF by default**, and it has to be transcribed, it is subjected to **positive regulation** by an **activator protein** bound to the **activator site in the promoter**.
- **The activator recruits the RNA polymerase** to initiate transcription.



Before the -35 – before the promotor

5. Regulation at the level of transcription initiation

5.3 Activator



An activator is going to bind upstream the promoter.

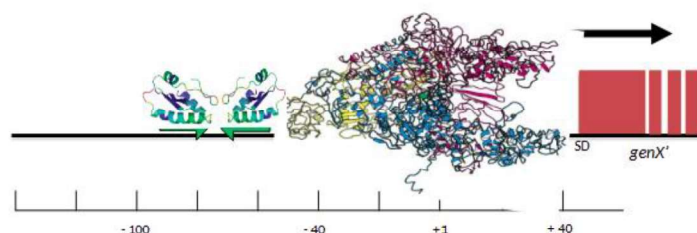
You have to know if an operator is overlapping or not and then you need to know where the activator is.

5. Regulation at the level of transcription initiation

5.3 Activator



In activation, an activator protein binds to an activator binding site to facilitate RNA polymerase binding - Hence simultaneous binding of both the activator and the RNA polymerase



Summary of repressor vs activator:

What we want to know: if something is an activator or repressor – you have to break it.

- If it was an activator, we see a decrease of the function
- if it was a repressor we see an increase of the function.

5. Regulation at the level of transcription initiation

Summary Repressor vs Activators



REPRESSOR

- The way repressors work is by preventing the binding of the RNA polymerase to the promoter. In order to work the binding sites of the repressor and the RNA polymerase must overlap
- Operators **overlap** with RNA polymerase binding site
- You could consider this as simple **competition** between the two proteins (repressor and the RNA polymerase)
- Exceptions do exist; they are however rare

ACTIVATOR

- The way activators work is by facilitating the binding of the RNA polymerase to the promoter. In order to work the binding sites of the activator and the RNA polymerase cannot overlap
- The **activator binding site is within the gene promoter (upstream) but does not overlap with the RNA polymerase binding site**
- **In the absence of the activator the RNA polymerase cannot bind** because the affinity for the promoter is low
- Exceptions do exist; they are however rare.

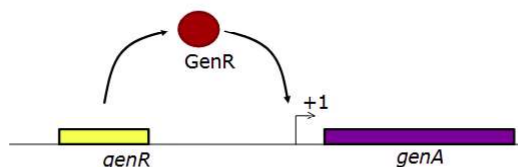
5.4 In the lab (mutants)

5. Regulation at the level of transcription initiation

In the lab



- How can we experimentally determine whether the expression of a given regulated gene is subject to repressor or activator control?
- We know that *genR* regulates *genA*, but don't know if positively or negatively
- **Break the elements systematically to discover the function** © (aka, make mutants)
- This goes always on the exam

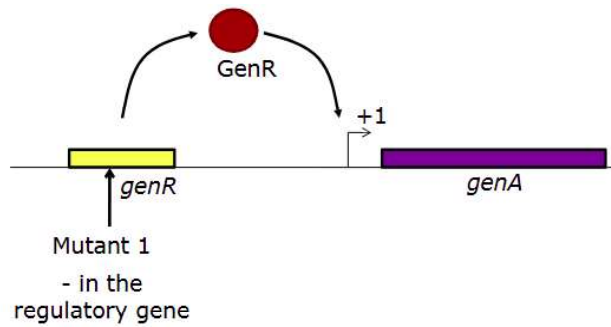


Question 1:

5. Regulation at the level of transcription initiation In the lab



- Let's assume I can measure precisely the concentration of genA product
- Mutant 1: in the regulatory gene.
 - How is the expression of genA if GenR is a repressor?
 - How is the expression of genA if GenR is an activator?



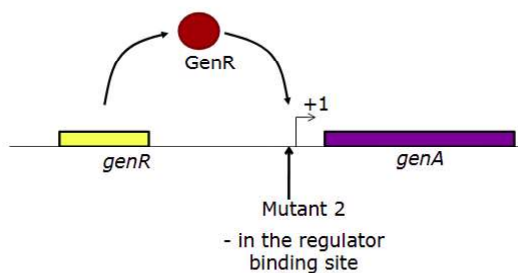
- Answer 1: the expression will be Higher
- Answer 2: the expression will be Lower

Question 2:

5. Regulation at the level of transcription initiation In the lab



- Let's assume I can measure precisely the concentration of genA product
- Mutant 2: in the regulatory binding site.
 - How is the expression of genA if GenR is a repressor?
 - How is the expression of genA if GenR is an activator?



- Answer 1: It goes ?
- Answer 2: It goes down

Regulation at the level of transcription initiation In the lab: repressor and activator

5. Regulation at the level of transcription initiation In the lab



REPRESSOR

- A loss of function mutation in the regulatory gene will lead to a constitutive high expression of the structural gene, regardless of the stimuli
- A mutation of the operator will lead to a constitutive high expression of the structural gene, regardless of the stimuli

ACTIVATOR

- A loss of function mutation in the regulatory gene will lead to a constitutive low expression of the structural gene, regardless of the stimuli
- A mutation of the activator site will lead to a constitutive low expression of the structural gene, regardless of the stimuli

5.5 Cis vs trans-acting

5. Regulation at the level of transcription initiation 5.4 Cis and trans-acting regulatory elements



Cis-Regulatory elements

- Non-coding DNA sequences in or near the genes they regulate. They serve as binding sites for transcription factors or other regulatory proteins
- They are physically linked to the genes they control (i.e., promoter)
- Example: light switch, directly connected to the bulb

Trans-regulatory elements

- Typically, proteins or RNAs that bind to cis-regulatory elements and influence transcription. They can be produced anywhere in the genome and act at multiple sites, not just the site (or gene) from which they were produced.
- They are diffusible, meaning, they can move around the cell
- Example: person that can switch on or off multiple bulbs



Cis-regulatory elements: Imagine that you have a gene that has a promoter and the promoter induce the expression of something but the promoter breaks, cannot be supplemented – it has to be physically linked to the genes they control

Trans-regulatory elements: can be supplemented and the functions can be restored (if the original breaks (sigma factors, repressor, activators))

How can you differentiate between a cis- and trans-regulatory element?:

- cis are usually promoters or operators
- trans are sigma factors, repressors, activators

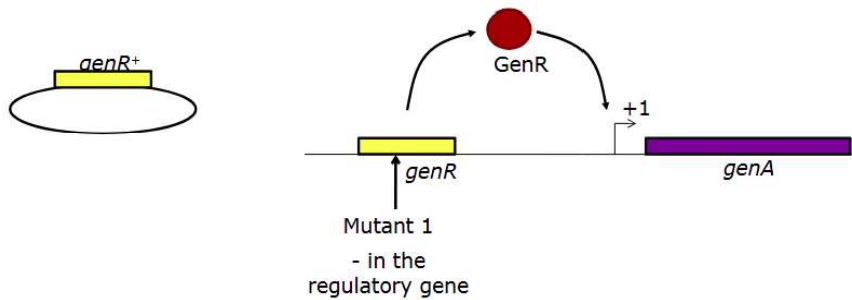
Back to our case – questions 3:



5. Regulation at the level of transcription initiation In the lab



- Let's assume *genR* is an activator of *geneR*
- Mutant 1: in the regulatory gene.
 - How is the expression of *genA* if *GenR* is an activator? – low
 - If we provide *genR* from an external source, would *genA* expression be restored?



We have to restore the broken bacteria (the broken *genR*)

- Answer 1: low
- Answer 2: yes

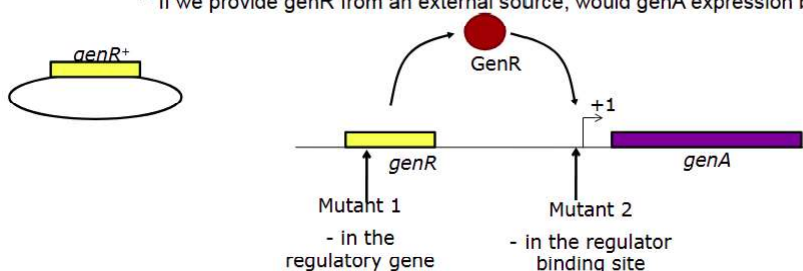
Question 4:



5. Regulation at the level of transcription initiation In the lab



- Let's assume *genR* is an activator of *geneR*
- Mutant 1: in the regulatory gene.
- Mutant 2: in the regulatory binding site.
 - How is the expression of *genA* if *GenR* is an activator? – low
 - If we provide *genR* from an external source, would *genA* expression be restored? No



You will not be able to differentiate whether you have broken the activator or the promoter – so you supplement.

“What am I breaking – how is it supposed to work if ?”

- Building an ethanol biosensor

Building an ethanol biosensor

Ethanol is a key metabolite in various industrial processes, including alcoholic fermentation and biofuel production. Detecting and quantitatively measuring its concentration in bioreactors can help optimize production and ensure quality control.

Objective: identify a promoter that responds to ethanol and ascertain the roles of various potential activator and inhibitor proteins in regulating this promoter's activity. You'll then conceive a synthetic biology strategy to build a biosensor for detecting ethanol in bioreactors.

AFU = arbitrary fluorescent units

Promoter	Regulatory Protein	GFP Expression (No Ethanol)	GFP Expression (With Ethanol)
P1	A	20 AFU	80 AFU
P1	B	20 AFU	10 AFU
P2	A	10 AFU	90 AFU
P2	B	10 AFU	100 AFU
P3	A	5 AFU	5 AFU
P3	B	5 AFU	5 AFU

- Which promoter(s) shows a significant change in GFP expression in response to ethanol?
- Which is the role of A and B on each promoter?
- Which promoter and which protein would you choose to build a biosensor?

We have three different promoters and two regulatory protein:

- Which promoter(s) shows a significant change in GFP expression in response to ethanol?

P2 and P1, though P1 has a greater background and less detection range

- Which is the role of A and B on each promoter?

In P1:

- A is the activator
- B is a repressor

In P2:

- Both A and B are activators
- Which promoter and which protein would you choose to build a biosensor?

P2 with the protein B is a strong candidate because it offers the lesser background and the highest dynamic range for measurement.

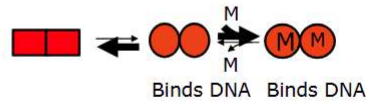
5.6 Inducing effector molecules

5. Regulation at the level of transcription initiation

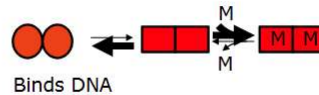
5.5 Inducing effector molecules



- The activators and repressors exist in two different conformations: The active form where they bind DNA and the inactive form where they cannot bind DNA



- The activators and repressors must be able to sense "something" that favors one of the conformations – stabilizing the least stable conformation



The most common thing is repressors and activators can bind to a metabolite and change the shape -and therefore their capacity for binding DNA changes too.

Activators can be an activator both when they are with and without a metabolite.

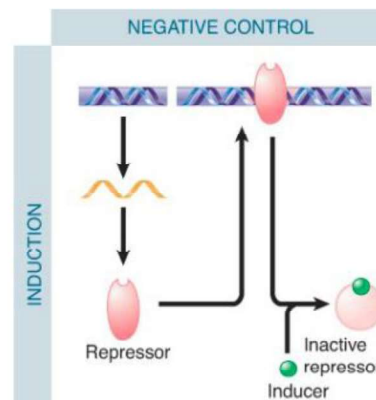
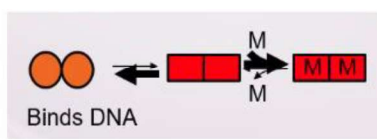
Inducible repression:

5. Regulation at the level of transcription initiation

5.5 Inducing effector molecules



- Inducible repression**
- A repressor is expressed and binds to the operator, avoiding the expression of a gene
- When the repressor is in the presence of an inducer, it adopts an inactive form that cannot bind the operator, allowing the expression of the associated gene



Repressor in the presence of an inducer makes the repressor inactive and thus it cannot bind the operator.

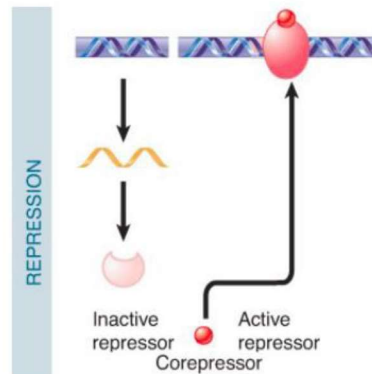
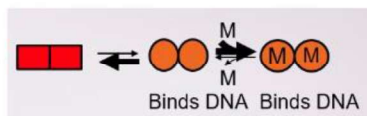
Co-repressor dependent repression:

5. Regulation at the level of transcription initiation

5.5 Inducing effector molecules



- **Co-repressor dependent repression**
- A repressor is expressed but cannot bind to the operator, so the controlled gene is expressed
- When the co-repressor is present, it binds to the repressor, enabling the complex binding to the operator and stopping the expression of the gene



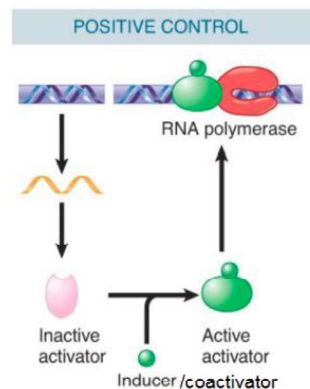
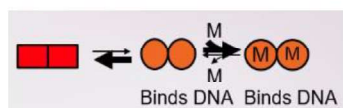
Inducible activation:

5. Regulation at the level of transcription initiation

5.5 Inducing effector molecules



- **Inducible activation**
- An activator is expressed but cannot bind upstream the promoter, so the controlled gene is off
- An inducer or co-activator can bind to the activator and allow the binding of the activator to recruit the RNA polymerase therefore expressing the controlled gene.



Repressible activation:

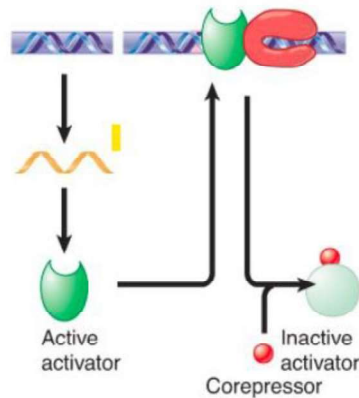
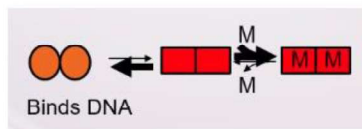
5. Regulation at the level of transcription initiation

5.5 Inducing effector molecules



- **Repressible activation**

- An activator is expressed and binds upstream to the promoter, so the controlled gene is on
- A co-repressor can bind to the activator and avoid the binding of the activator to the promoter turning off the controlled gene.



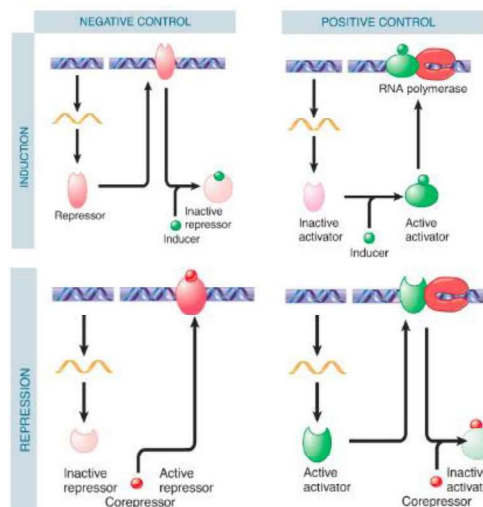
Summary of the inducing effector molecules:

5. Regulation at the level of transcription initiation

5.5 Inducing effector molecules



Role	Nature	Procedence
Repressor	Protein	Gene encoded
Activator	Protein	Gene encoded
Co-repressor	Metabolite	Environmental
Co-activator	Metabolite	Environmental
Inducer	Metabolite	Environmental



The metabolites is the signals that comes from outside the cells and tells them that something is going on and it needs to change. The proteins is the ones in contact with the DNA.

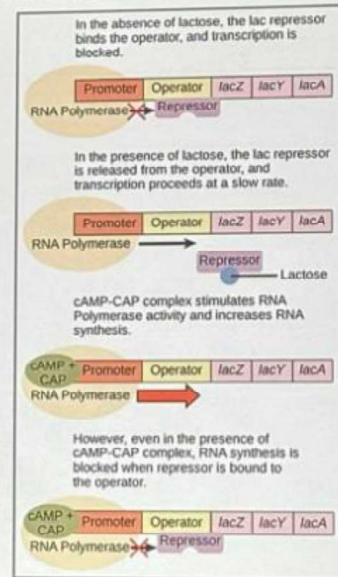
6. The lac operon

6.1 Repressing expression by the Lac repressor

Lac operon + catabolyte repression = a repressor must leave and an activator must arrive

- A operon is a collection of genes transcribed in the same piece of mRNA in the genome of prokaryotes.
- The lac operon expresses genes involved in the metabolism of lactose as carbon source
- The preferred carbon source for bacteria is glucose, therefore to open the lac operon:
 - Lactose must be present (removes a repressor)
 - Glucose must be active (attracts an activator)

<https://www.youtube.com/watch?v=bibDhYVPiIc>

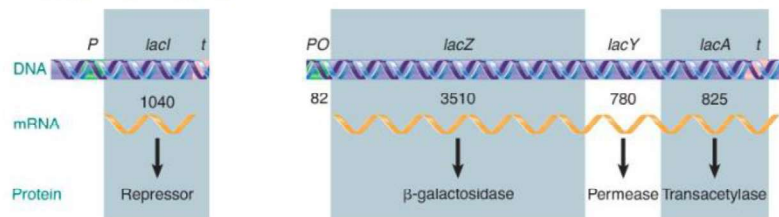


6. The *lac* operon

6.1 Repressing expression by the Lac repressor



- Bacteria LOVES glucose. More than any other carbon source available. Therefore, the machinery to eat glucose is always there.
- However, bacteria must adapt to other carbon sources to survive, and therefore they can induce the expression of specific transporters and enzymes to metabolize other carbon sources (such as lactose)
- Genes coding for proteins that function in the same pathway may be located adjacent to one another and controlled as a single unit that is transcribed into a polycistronic mRNA – they constitute an **operon**



If there is glucose available the system is perfectly able to use glucose and nothing else. Only when glucose is not available but other sugars are, then the metabolism will put machinery to use – like the lac-operon. (which is a group of proteins that makes a machinery which makes it possible to “eat” lactose.

- High glucose low lactose

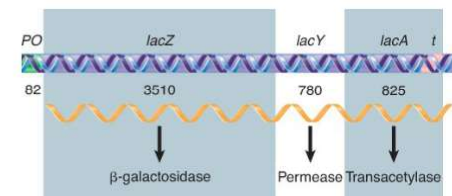
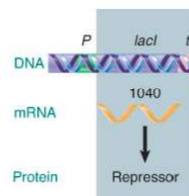
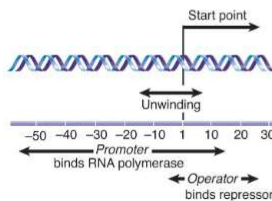


6. The *lac* operon

6.1 Repressing expression by the Lac repressor



- In normal conditions (high glucose and **low lactose** available)
 - The expression of *lacZYA* is very low (basal) because its expression is **inhibited** by a **repressor protein (LacI)**
 - There is some beta-galactosidase, permease and transacetylase
 - LacI binds to the **lactose operator**, which overlaps with the RNA polymerase binding site



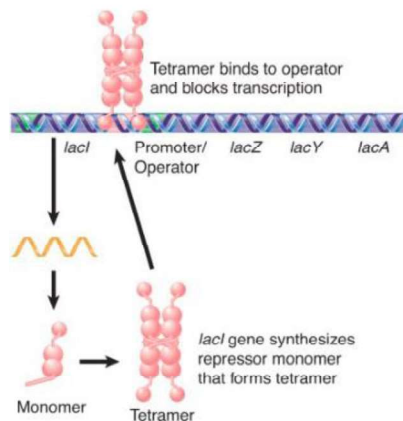
It looks more or less like this:

6. The *lac* operon

6.1 Repressing expression by the Lac repressor



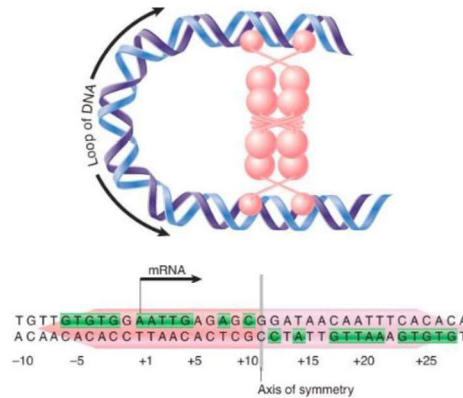
- *lacI* is constitutively expressed (LacI)
- LacI tetramerizes and binds to the operator in the absence of lactose
- *LacZYA* cannot be expressed because RNA polymerase cannot bind to the promoter



6. The *lac* operon

6.1 Repressing expression by the Lac repressor

- The gene *lacI* is constitutively expressed, producing the protein LacI
- LacI tetramerizes and binds to the operator multiple operators sites in the absence of lactose
- This cause the DNA to bend making it even more complicated for the RNA polymerase to access the promoter
- LacZYA cannot be expressed because RNA polymerase cannot bind or even access to the promoter
- The operator is a palindromic sequence of 26 bp.



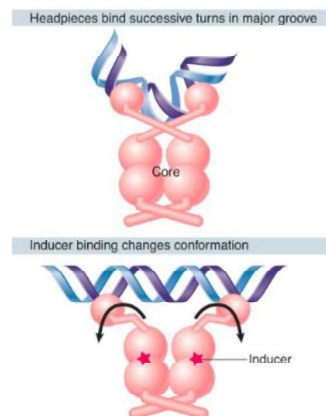
- What happens when lactose is available: low glucose high lactose



6. The *lac* operon

6.1 Repressing expression by the Lac repressor

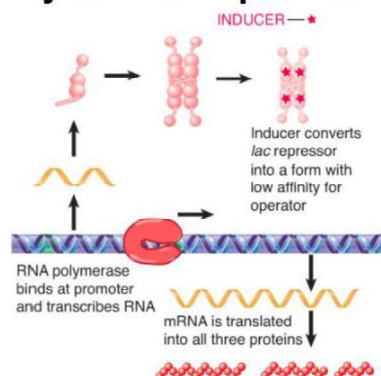
- In induction conditions (low glucose and high lactose available)
 - β -galactosidase works at V_{max} and starts accumulating allolactose
 - Allolactose interacts with LacI, transforming it into a protein form with decreased affinity for the operator
 - LacI is released from the **lactose operator**, which overlaps with the RNA polymerase binding site
 - The expression of *lacZYA* is high because is expression is **induced** by an inducer metabolite (lactose)



6. The *lac* operon

6.1 Repressing expression by the Lac repressor

- An inducer functions by converting the repressor protein into a form with lower operator affinity.
- The *lac* repressor is inactivated by an allosteric interaction in which binding the inducer at its site changes the properties of the DNA-binding site.



6.2 Activating expression by cAMP-CRP

Catabolite repression, catabolite repressor protein (CRP) carbong catabolite contron protein A (CcpA) :

6. The *lac* operon

6.2 Activating expression by cAMP-CRP

- **Bacteria LOVES glucose.** More than any other carbon source available. Therefore, the machinery to eat glucose is always there. And if enough glucose is available, not matter how much is available from other sources, the cell will shut down other alternative routes to consume preferably glucose.
- The lactose operon is regulated by a second layer of control: catabolite repression
- **Catabolite repression:** global network that consists of the ability of glucose to prevent the expression of a number of genes (in *E. coli* >20 operons)
- **Catabolite repressor protein (CRP)** is an activator protein that binds to a target sequence at a promoter in many gram-negative bacteria. (i.e., *E. coli*)
- **Carbon catabolite control protein A (CcpA)** works as a repressor for genes encoding enzymes used for the utilization of alternative carbon sources in many gram-positive bacteria. (i.e., *B. subtilis*)



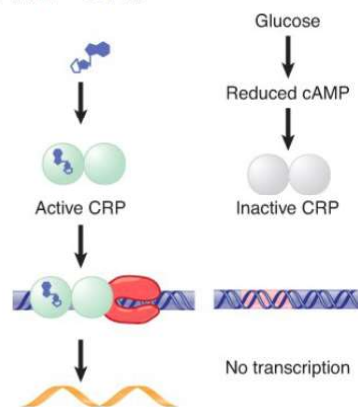
Catabolite repressor protein (CRP), cAMP : Lac operon induction : glucose lo+lactose high



6. The *lac* operon

6.2 Activating expression by cAMP-CRP

- **Catabolite repressor protein (CRP):** activator protein
- **cAMP:** inducer (metabolite)
- The concentration of Glucose is inversely related to the concentration of cAMP (High glucose → low cAMP, low glucose → high cAMP)
- When glucose is low, cAMP is high, then it binds to CRP, CRP dimerize, then the dimer binds to the promoter recruiting the RNA polymerase which will transcribe the lac operon structural genes



Lac operon induction: glucose low + lactose high

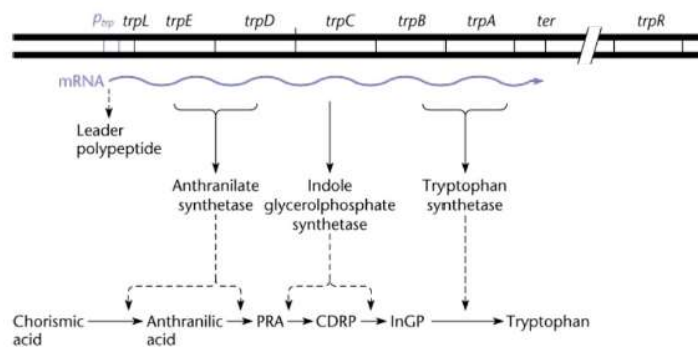
The repressor has to leave (eliminated from the DNA) (the repressor of lactose) and the activator has to arrive because of the absence of glucose. – when both thing happens that's when this happens.

7. The Trp operon

7. The *Trp* operon

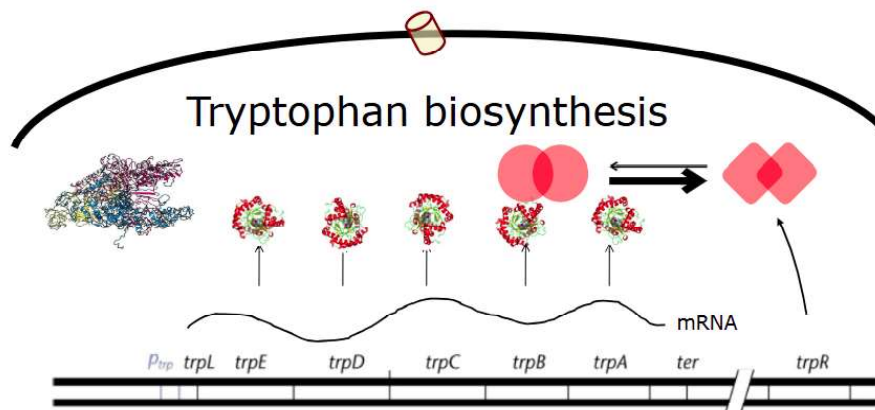


- The *trp* operon is negatively controlled by the level of its product, the amino acid tryptophan (autoregulation).
- The regulation is at two levels – repressor regulation at transcription initiation and through a terminator/antiterminator mechanism (Regulatory RNA class)



Tryptophan has two regulatory levels: repressor regulation at transcription initiation and through a terminator /anti terminator mechanism

7. The *Trp* operon



Only happens in the absence of tryptophan!!

In the absence of lactose and the presence of glucose, what happens to the Lac operon in *E. coli*?

The repressor protein (LacI) has a low affinity for the operator

0%

The structural genes of the Lac operon are transcribed and translated

0%

The repressor protein (LacI) binds with high affinity to the operator, preventing translation

0%

The catabolite repressor protein (CRP) binds cAMP and binds with high affinity to the operator, preventing translation

0%

- Answer: The repressor protein (LacI) binds with high affinity to the operator, preventing translation

You suspect a protein (A) is regulating the expression of another protein (B) in E.coli. The expression of B is normally very low. You mutate A and the expression of B is suddenly very high. What is the role of A in the regulation of the expression of B

Activator

0%

Repressor

0%

Inducer

0%

Co-repressor

0%

- Answer: Repressor

You have noticed that the presence of a metabolite modifies the levels of expression of protein A. You are trying to identify the gene encoding the regulator that can sense the metabolite and have narrowed it down to 3 candidates: genes X, Y, and Z. From the table, deduce which is the regulator gene and what kind of regulation it is.

Metabolite present (Y/N)	Mutated gene	Level of A (100)
Y	-	0
N	-	100
Y	Gene X	0
Y	Gene Y	100
Y	Gene Z	0

Gene X, co-repressor dependent repressor		##.##%
Gene Y, repressor		##.##%
Gene Y, co-repressor dependent repressor		##.##%
Gene Y, inducible activator		##.##%

- Answer: Gene Y, co-repressor dependent repressor

Notes for the videos:

- “Chromatin remodeling in eukaryotes”

Chromatin remodeling is the process that starts the transcription in eukaryotes. DNA is wrapped around proteins called histones. DNA has a sugar phosphate backbone and because of that DNA has a negative charge due to the phosphate groups and the histones carry a positive charge. This makes sense as opposite charges attract. Histones are proteins which means that they are made from long chains of amino acids, meaning histones positive charge is coming from the amino acids that it is built from. The amino acids that have positive charges are the basic ones: Histidine, arginine and lysine. Histones surface is rich in lysine residues, so they have a positive charge on their surface, where they interact with DNA. In order for transcription to occur we cannot have the histones in the way, the DNA needs to be “free” so that the RNA polymerase and any transcription factors can go in, find a promoter region, bind to it and start transcribing. For this there exists an enzyme called Histone Acetyl Transferase (HAT) which removes the positive charged lysine residues by adding an acetyl group to the lysine from the histones (making the charge neutral) removing the positive charge and thereby freeing the DNA as the histone affinity to the DNA has decreased. When the DNA is unpacked it is ready for transcription.

If you want to repack the DNA – so go backwards from what just has happened. For this there is an enzyme called “Histone Deacetylases” (HDAC’s). They deacetylate the histones – get rid of the acetyl group and reestablish the positive charge of the histones (made by the lysine residue on the surface of the histones) so you can have the DNA reassociate with the histones.

- “DNA Binding motifs overview”

In order for transcription to start we need the minimal requirement of general transcription factors and RNA polymerase. Apart from the general transcription factors there are other

specialized transcription factors which bind specific regions on the DNA and recruits activator or repressor and thereby regulating the transcription.

How a transcription factor binds to the DNA – it turns out that in transcription factor there are specific DNA binding motifs which ensures the specificity of that specialized transcription factor to bind in a sequence specific manner – and thereby activating or depressing transcription.

Molecular Biology Week 6: Regulation of gene expression in eukaryotes

Learning objectives

After today's lecture and exercises, a student who has met the objectives will be able to:

- Describe the overall process of how a gene is expressed in a eukaryote
- Explain the modular nature of DNA-binding proteins
- Describe the terms Epigenetics, Boundary element, Repressor, Activator, True activator, Antirepressor, Yeast-two-hybrid assay, Basal apparatus, Chromatic remodelling complex
- Describe the modular structure of regulatory proteins (DNA-binding domain and Transcription-activation domain)
- Interpret a simple experiment with the yeast-two-hybrid assay
- Describe how a Chromatic remodelling complex can remove histones from the DNA, and how histone modification (eg. Acetylation) controls chromatic remodelling.
- Describe how DNA modification or the assembly of proteins on DNA causes epigenetic effects

Notes from the lecture

1. Introduction

Angelina Jolie<3

Why did Angelina Jolie undergo a double mastectomy?

- Angelina had a mutation in the regulatory gene BRCA1.
- BRCA1 is a transcription factor that controls cell cycle, cell replication and DNA repair.
- When overexpressed, it causes apoptosis
- When mutated, cells grow without control causing aggressive tumors
- Mutations in BRCA1 increase the risk of breast cancer up to 72%.



- Her mutation causes a gain of function – cell growth is out of control
- For now, the only way to treat this is to remove the organ that is affected – a double mastectomy.

a) Why regulating gen expression?



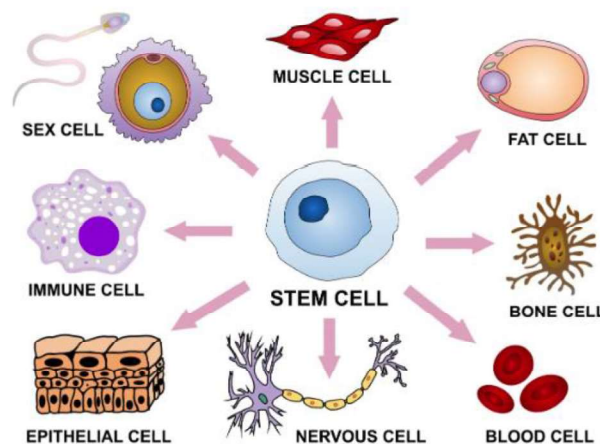
1. Introduction

a) Why regulating expression

- Cell differentiation

All the cells in a pluricellular organism have the **same genetic content (DNA)**.

Tissue differentiation and therefore specialization depends on the **differential expression of genes (mRNA)**.



- Stem cells, cell differentiation
- In prokaryotes the regulation of gene expression is mainly about survival and the need to adapt to the environment
- For eukaryotic gene expression it is needed as all cells start from a stem cell. So all cells in a pluricellular organism (fx humans) have the same genetic content (DNA)

- We also need it as an environmental response.

Central dogma (protein synthesis) for eukaryotes :

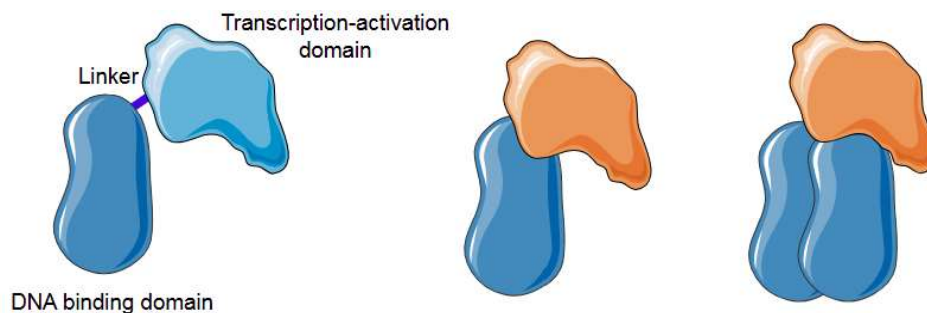
- For prokaryotes the whole central dogma happens in the cytoplasm but for the eukaryotes the DNA is firstly unpacked from the histones (chromatin removal) chromatin remodelling is the main expression regulation checkpoint in eukaryotes – this makes chromatin remodeling the MOST important step for protein synthesis in eukaryotes.
- Transcription (pre mRNA), splicing and maturation (mature mRNA) also happens in the nucleus as well as the chromatin remodelling.
- Translocation is the next step where the mature mRNA is transported to the cytoplasm where translation happens.

Transcription factor: DNA binding proteins (eukaryotes)

1. Introduction

c) Transcription factor: DNA binding proteins

- DNA binding and transcription-activation activities are carried out by independent domains of an activator
- The role of the DNA-binding domain is bringing the transcription-activation domain into the vicinity of the promoter



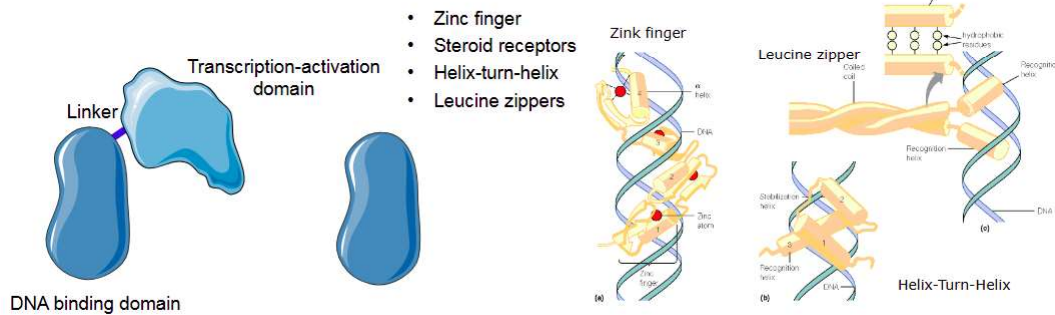
- The DNA binding domain and the transcription-activation domain can be too “separate” or “different” proteins. These bind to each other by a linker – binding to the DNA binding protein in order to interact with the DNA.

Activator structure:

1. Introduction

c) Activator structure: DNA binding proteins

- Activators are classified according to the type of DNA-binding domain
- Members of the same group have sequence variations of a specific motif that confer specificity for individual DNA target sites



- Zinc finger, steroid receptors, helix-turn-helix, leucine zippers

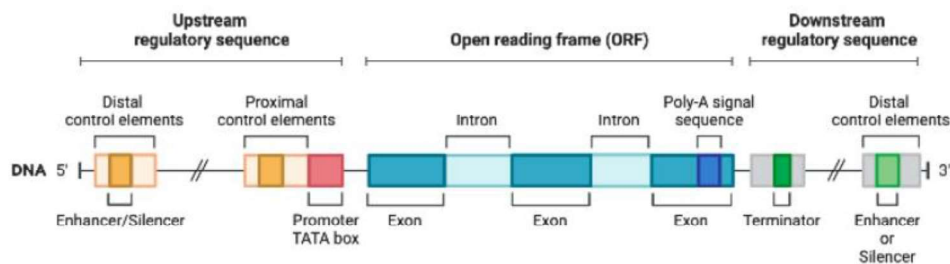
Repressors in a prokaryotic gene would bind around the -35 and the -10 part (it competes with the RNA polymerase which also binds here). An activator binds upstream the promoter (the -35 and -10) so further to the left.

Eukaryotic promoters are way more complicated than prokaryotic promoters.

- Eukaryotic gene structure:

2. Repetition: The Start Point for RNA Polymerase II

Eukaryotic Gene Structure

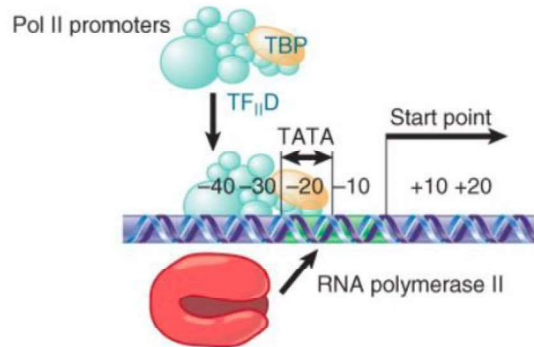


The basal apparatus assembles at the RNA pol II promoters in eukaryotic protein synthesis (transcription):

2. Repetition: The Start Point for RNA Polymerase II

The Basal Apparatus Assembles at the RNA Pol II Promoters

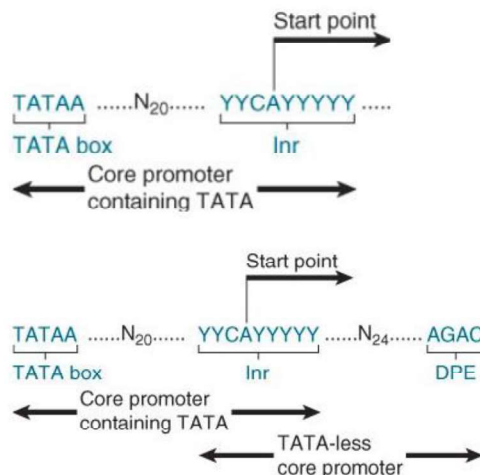
- TBP is an universal factor necessary for all polymerases to bind their promoters
- The recruiter factor for RNA polymerase II is **TFIID**, which consists of TBP and about 14 TBP-associated factors (**TAFs**), with a total mass of about 800 kD.
- Binding of TFIID to the TATA box or Inr is the first step in initiation.



2. Repetition: The Start Point for RNA Polymerase II

RNA polymerase II synthesizes mRNA in the nucleoplasm.

- A core promoter for RNA polymerase II includes the **Inr** and, commonly, either a **TATA box** or a **DPE**.
- **TATA-containing** or **TATA-less** promoters: The **downstream promoter element (DPE)** is a common component of RNA polymerase II promoters that do not contain a TATA box (**TATA-less** promoters).



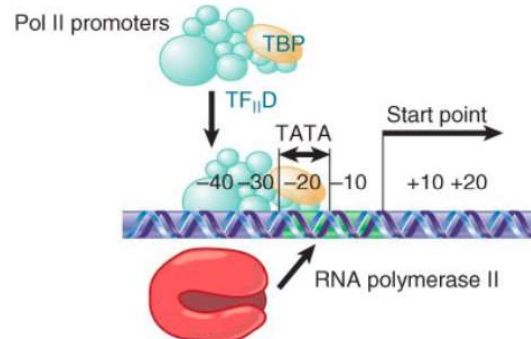
- The three kind of promotors that you can have in eukaryotic transcription: TATA box, downstream promotor element and there is a last one??

2. Repetition: The Start Point for RNA Polymerase II



The Basal Apparatus Assembles at the RNA Pol II Promoters

- TBP is an universal factor necessary for all polymerases to bind their promoters
- The factor for RNA polymerase II is **TFIID**, which consists of TBP and about 14 TBP-associated factors (**TAFs**), with a total mass of about 800 kD.
- Binding of TFIID to the TATA box or Inr is the first step in initiation.
- **The upstream elements and the factors that bind to them increase the frequency of initiation.**
- Regulation of gene expression!



Which is the main point of gene expression regulation on eukaryotes

The recruitment of activators to the promoter

0%

Chromatin remodelling

0%

mRNA maturation

0%

Repression mediated by metabolites

0%

- ANSWER: Chromatin remodeling

All the eukaryotic genes have a TATA box which is indispensable for transcription initiation

- ANSWER: FALSE
- You always need the TATA binding protein, but not the tata box, if there is no promotor it will bind to the Inr.

b) Targets for regulating gene expression

c) Activator structure: DNA binding proteins

2. Repetition: the transcription initiation process in prokaryotes

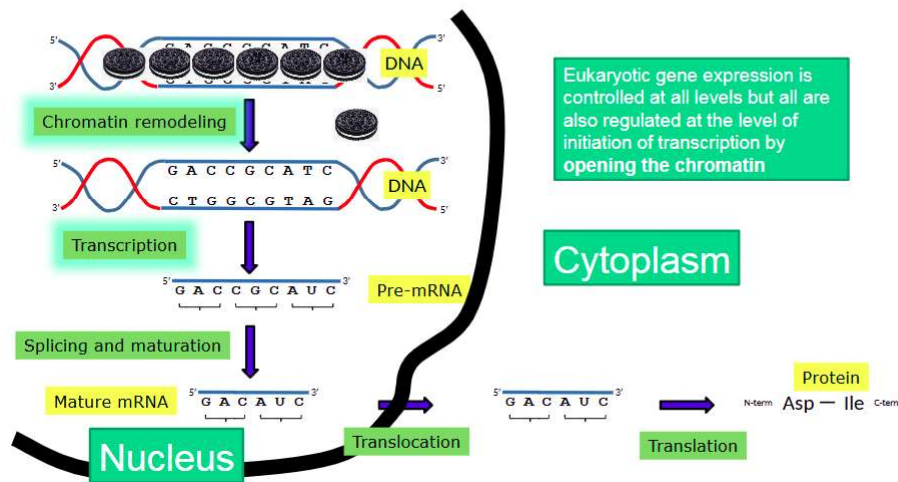
3. Repetition: The Start Point for RNA Polymerase II

Eukaryotic gene expression

3.0 Overview

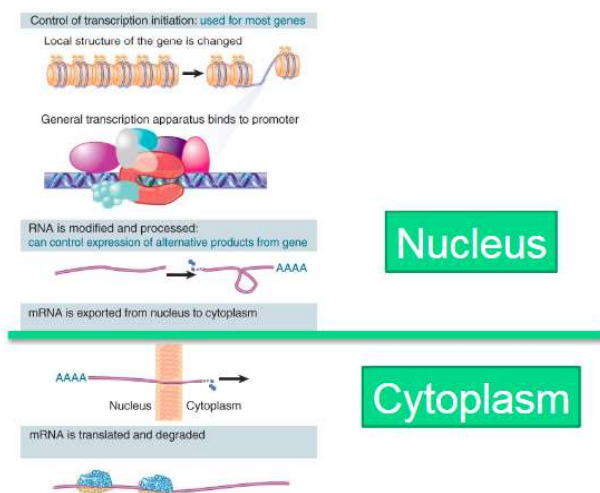
- In this lecture we are looking at the chromatin remodelling and transcription

3. Overview: How to turn on a gene



How to turn on a gene

3. Overview: How to turn on a gene

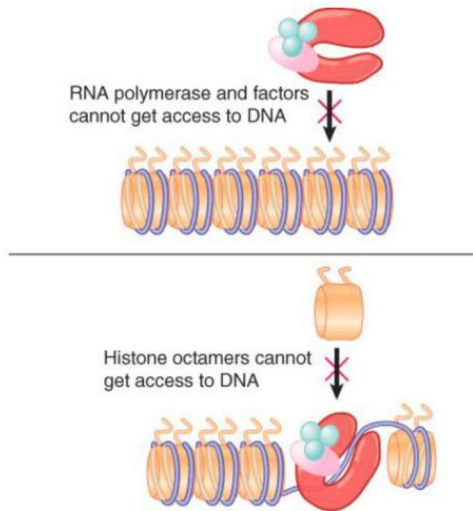


1. Chromatin opens by the action of histone remodelling enzymes* **
2. Transcription factors are bound to enhancer regions by direct contact with the DNA*
3. Transcription-activator domains recruit the transcription apparatus by direct interactions with their elements
4. The RNA pol II transcribes a gene into mRNA
5. mRNA is processed by splicing
6. The mature mRNA is translocated to the cytoplasm
7. mRNA is translated and degraded

- Activators can recognize their targets in a closed chromatin so that they can recruit the ones that are going to open the chromatin (packed DNA in histones)
- The genome is divided into domains by boundary elements (insulators) – insulators can block the spreading of chromatin modifications from one domain to another – they are tightly bound to the DNA, and they block the whole DNA from being unpacked – so only the needed domain gets unpacked – they are indispensable for life

4. Chromatin remodeling

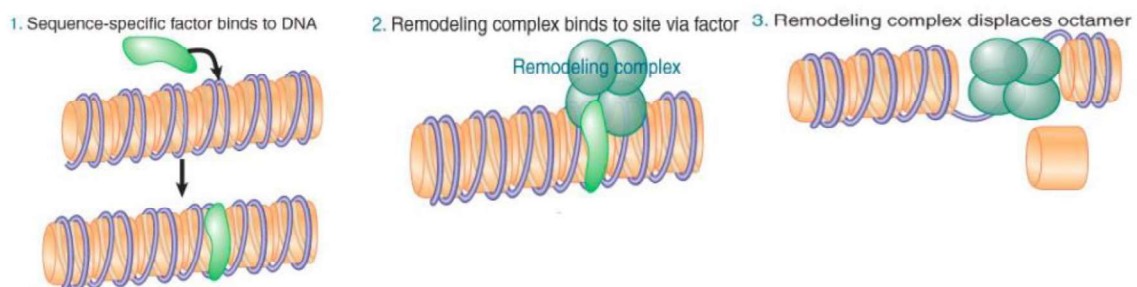
4. Chromatine remodeling: overview



- The RNA pol cannot access to DNA when DNA is packed around histones

- Heterochromatin?
- Remodeling complex carries out the active chromatin remodeling process – some remodeling complexes can exchange one histone by other in the nucleosomes
- All modeling complexes contain an ATPase catalytic subunits – they can alter, lside og even displace nucleosomes
- The energy needed in the process comes from the hydrolysis of ATP.

4. Chromatine remodeling : overview

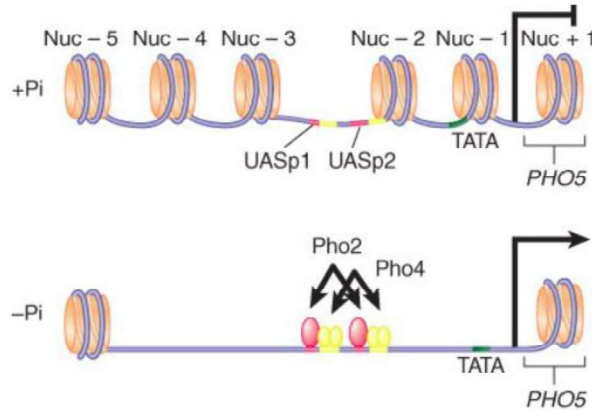


1 *Some transcription factors can recognize their targets DNA sequences in closed chromatin to initiate activation.

2. The activator recruits Chromatin remodelling complex

3. Due to the action of the complex, the nucleosome is displaced, leaving the promoter sequence reachable to transcription factors and the RNA pol

4. Chromatine remodeling: example PHO5



Pho2 and Pho4: transcription activators

UASp1 and UASp2: DNA enhancer sequences of the PHO5 gene

PHO5: gene encoding for a excreted phosphatase

Pho2 and Pho4 transcription activators responds to phosphate starvation. They activate, translocate into the nucleus and bind to their target sequences.

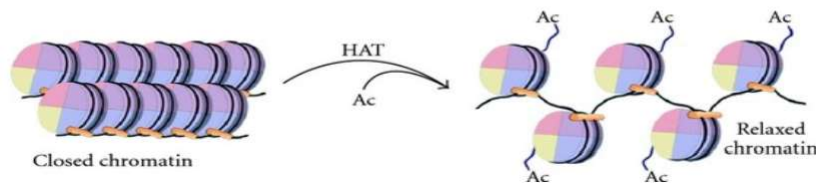
They recruit chromatin-remodelling complexes that displace the nucleosome leaving the promoter and the TATA box free for RNA-pol II binding, allowing PHO transcription

- Pho2 and ph o4 are activators that the absence of the metabolite. They recruit the chromatin remodeling complexes that displace the nucleosome leaving the promotor at the TATA box free for RNA-polymerase II binding – allowing PHO transcription.
- Some components of the transcriptional apparatus work by changing chromatin structure
- Activation is associated with a looser chromatin structure
- Repression is achieved by stabilizing a dense chromatin structure or by binding to and masking activators that otherwise would induce a looser chromatin structure
- Acetylation is always going to be related to activation and de-acetylation is always going to be related to repression.

Histone acetylation is associated with transcription activation

4. Chromatine remodeling

Histone Acetylation Is Associated with Transcription Activation

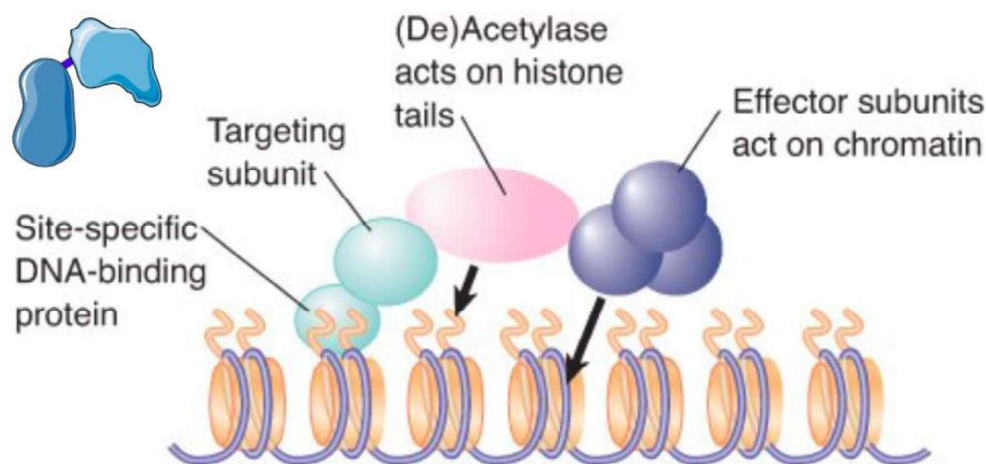


- **Histone acetylation** is associated with **activation of gene expression**.
- **Histone acetyltransferases (HAT)** – Enzymes typically associated with transcription activators present in large complexes that acetylates **lysine residues** in histones.
- The **net charge** of the histones changes **from + to 0**, losing its **electrostatic interaction with DNA**

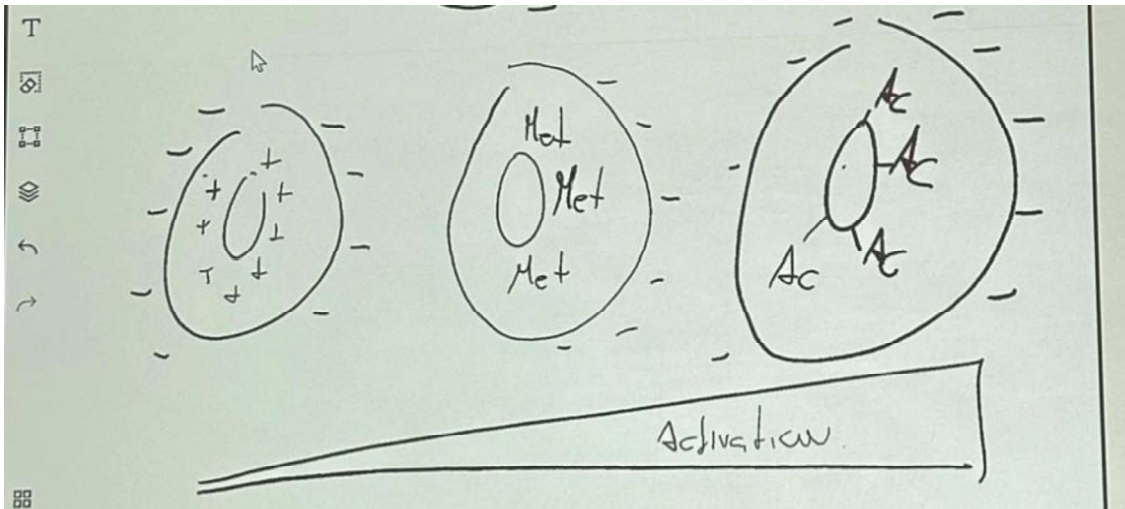
DTU

4. Chromatine remodeling

Histone Acetylation Is Associated with Transcription Activation

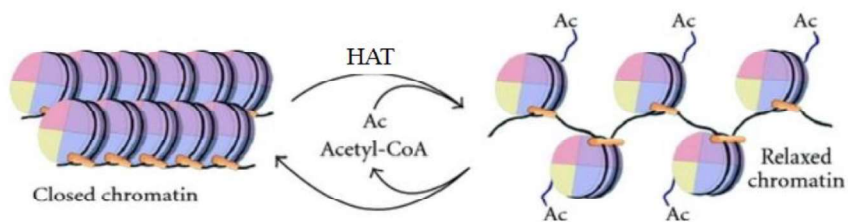


- As we see the system is very complex because of binding of multiple proteins
- Acetylation is always going to be related to activation and de-acetylation is always going to be related to repression.
- Methylation can mean both activation or repression depending on the starting point. The DNA will stay closer to the histones than when use of the acetylation as there still will be a partial negative charge (from the histones). So with a non-acetylated histones (you have nothing), methylation would mean activation. But if the histones are already acetylated, methylation would be a form of repression.



4. Chromatine remodeling

Histone De-acetylation Is Associated with Transcription Repression



- **Histone de-acetylation** is associated with **repression** of gene expression.
- **Histone deacetylase (HDAC)** remove acetyl groups from lysine residues on histones; may be associated with repressors of transcription. **HDACs are inhibited by butyric acid**
- The **net charge** of the histones changes **from 0 to +**, increasing its electrostatic interaction with **DNA**

This picture is showing repression: